

Comprehensive Errata and Rectification Ledger for the Post-Mitigated Abraxas Model (PMM)

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Abstract

This document serves as the formal rectification ledger for the Post-Mitigated Abraxas Model (PMM) and the APU-X neuromorphic substrate thesis. It systematically catalogs 17 issues identified during peer review—ranging from critical mathematical sign errors to missing bridging lemmas and unsourced hardware parameters. This ledger provides the explicit corrections, updated derivations, and structural implementations required to resolve these vulnerabilities prior to the final compilation of the unified thesis.

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1 Critical Errors

1.1 Issue 1: Sign Error in Euler-Lagrange Equation

Location: Definition 1.3 (Explicit Euler-Lagrange Guided Subspace).

Problem: The objective is to maximize $\text{Tr}(W^\top CW) + \alpha \cdot \text{Tr}(W^\top GG^\top W)$. Taking the derivative and setting it equal to the Lagrange multiplier terms yields a positive sign, not a negative one. A minus sign implies regularization opposes margin gain, contradicting the core theorem.

Correction: Change all instances of $(C - \alpha GG^\top)W^*$ to $(C + \alpha GG^\top)W^*$.

1.2 Issue 2: Truncated Proof

Location: Constant-Complete GAS-ECF Margin Gain Theorem.

Problem: The proof ends abruptly after declaring a Sylvester equation.

Correction: Insert the complete proof steps, specifically:

1. The Sylvester solution formula.
2. The Frobenius norm bound $\|W^{(1)}\|_F \leq \frac{\|V_\perp^\top GG^\top V_k\|_F}{\delta_k}$.
3. The explicit margin expansion demonstrating linear gain vs. quadratic penalty.

1.3 Issue 3: Numerical Arithmetic Error in Thermal Invariant

Location: Isotropic Thermal Dissipation Invariant.

Problem: The calculated value for ΔT_{max} was off by a factor of 1,000 due to a unit conversion error.

Correction: Update the calculation:

$$\Delta T_{max} = \frac{(12.5 \times 10^9) \times (150 \times 10^{-9})^2}{2 \times 1400} = \frac{2.8125 \times 10^{-4}}{2800} \approx 1.004 \times 10^{-7} \text{ K}$$

Update the follow-through bound: $\epsilon_{arith} \leq (10^{-5}) \times (1.004 \times 10^{-7}) = 1.004 \times 10^{-12} \ll 10^{-3}$.

1.4 Issue 4: Inconsistent Asymptotic Bound in Quicksand Oja++

Location: Constant-Explicit Event-Driven Subspace Tracking.

Problem: The theorem statement claims $\mathcal{O}(\eta_t B^2 / \delta_k^2)$, but the proof yields $\mathcal{O}(\eta_t B^4 / \delta_k)$.

Correction: Standardize the theorem statement to match the derived mathematical proof: “converges uniformly to an asymptotic bound proportional to $\mathcal{O}(\eta_t B^4 / \delta_k) + \mathcal{O}(\sigma_{noise}^2)$ ”.

2 Consistency Conflicts

2.1 Issue 5: Conflicting Quantization Noise Tolerance

Location: Robustness to Non-PSD Quantization Noise.

Problem: Different documents state conflicting conditions ($\|\mathcal{E}_{noise}\|_2 < \lambda_k$ vs. $\|\mathcal{E}_{noise}\|_2 \leq \frac{1}{2}(\lambda_k - \lambda_{k+1})$).

Correction: Unify the bound to the mathematically proven condition: $\|\mathcal{E}_{noise}\|_2 < \lambda_k$. Add a remark clarifying that this is the necessary permissive condition for preserving submodularity.

2.2 Issue 6: Two Different Forms of $L_{contract}$

Location: GIM-Induced Contraction Theorem.

Problem: Inconsistency between using a static η_t vs. adaptive η_t^{eff} , and $\mathcal{O}(\eta_t^2)$ vs. explicit B^4 .

Correction: Standardize to the adaptive step-size formulation. Add the following remark:

Remark: In the presence of adaptive step-size η_t^{eff} , the contraction modulus becomes $L_{contract} = 1 - 2\eta_t^{eff}\delta_k + (\eta_t^{eff})^2\kappa(\hat{\sigma}_t^2)$. Under the noise-adaptive regime, the quadratic term is uniformly bounded by $\eta_t^{eff} \cdot \delta_{min}$, maintaining $L_{contract} < 1$.

2.3 Issue 7: Allocation Probability Model ($\mathcal{O}(\log t)$ Bound)

Location: Hierarchical Memory Compression Theorem.

Problem: The transition from margin constraints to an allocation probability of α_0/t is assumed, not derived.

Correction: Insert a **Lemma (Covering Number for Margin Allocation)** before the theorem. Justify that if the underlying data manifold has a bounded intrinsic dimension d_{intr} , the covering number grows linearly $N_{cover} \propto t$, resulting in the allocation probability $P(I_{t+1} = 1|M(t)) = \alpha_0/t$.

3 Incomplete Proofs & Missing Justifications

3.1 Issue 8: Necessity Argument in GIM Theorems

Location: Global-to-Local Validity / GIM Mandatory Necessity.

Problem: The proof conflates a boolean verifier failure ($V_i = 0$) with immediate unbounded trajectory divergence.

Correction: Rewrite the Necessity ($\implies$) proof by contradiction: Assume $V_{global} = 0$ while the trajectory remains bounded. A global failure implies the unattenuated gradient field has diverged, but a bounded trajectory requires a bounded gradient field (continuity)—a contradiction. For local verifiers, note that $V_i = 0$ allows subsystem i to accumulate unbounded error independent of global monitoring due to cross-domain interactions.

3.2 Issue 9: Boundary Behavior of Adaptive Step-Size

Location: Transforming the Singularity / Barrier Lyapunov Function.

Problem: The mechanism driving $\lim_{\hat{\delta}_k \rightarrow \delta_{min}^+} \eta_t^{eff} = 0$ is not formally linked to the noise estimate $\hat{\sigma}_t^2$.

Correction: Add a **Lemma (Adaptive Step-Size Damping at Boundaries)**. Show that as $\hat{\delta}_k \rightarrow \delta_{min}$, noisy directions dominate the spectrum, causing $\hat{\sigma}_t^2 \approx \sum_{j>k} \hat{\lambda}_j(t)$ to increase towards infinity. Thus, $\lim_{\hat{\sigma}_t^2 \rightarrow \infty} \eta_t / (1 + \hat{\sigma}_t^2 / \delta_{min}) = 0$.

3.3 Issue 10: 95% Reconstruction Accuracy Figure

Location: Memory Reconstruction.

Problem: A 95% accuracy figure is stated as a theorem result, but it is actually an experimental specification.

Correction: Rewrite the theorem to solely state the geometric condition: $\epsilon_d \leq (\gamma_{margin} - \|\Delta x\|_2) / (\gamma_{margin} + \|\Delta x\|_2)$. Move the 95% figure strictly to the Experimental Protocols section as a validation target.

3.4 Issue 11: CHS Maxwell Boundary Condition

Location: CHS Electromagnetic Decoupling Proof.

Problem: The proof jumps abruptly from the idealized continuous limit ($A_{magnetic} \equiv 0$) to a physical leakage coefficient δ_{shield} .

Correction: Explicitly separate the proof into two phases: **Ideal Case** (rigorous derivation of $A_{magnetic} \equiv 0$) and **Physical Realization** (modeling finite graphene conductivity as $A_{physical} = \delta_{shield} \cdot A_{ideal}$).

3.5 Issue 12: Phonon Relaxation Time & Arithmetic Precision

Location: Acoustic Phonon Relaxation via Graphene/h-BN.

Problem: The connection between phonon scattering and the 10^{-3} precision bound is asserted, not derived.

Correction: Add a **Lemma (Phonon-Limited Thermal Transient)**. Formulate the transient temperature rise and show that enforcing $f_{SOT} < 1/\tau_{phonon}$ allows full scattering, bounding the peak thermal spike: $\Delta T_{peak} \leq \delta_{thermal} \cdot P_{density} \cdot d_{layer}/\kappa_{vertical}$, which guarantees $\epsilon_{arith} < 10^{-3}$.

3.6 Issue 13: MLRP Routing Lemma Integration

Location: Piecewise-MLRP Threshold Lemma.

Problem: The architectural application of the lemma is missing.

Correction: Add an **Application in PMM** subsection. Clarify that this policy is used by the Symbolic Reasoning Core (SRC) when routing discrete states between inference pathways, ensuring deterministic mode transitions based on ECF confidence metrics.

4 Minor Clarifications Needed

4.1 Issue 14: Additive vs. Multiplicative Failure Bound

Location: Explicit Hierarchical Error Bound.

Problem: The text says “bounded multiplicatively” but the formula is a summation (additive union bound).

Correction: Change the text to “bounded by the sum of independent conditional failure rates (additive bound)”.

4.2 Issue 15: Hardware Parameters Not Sourced

Location: Microarchitectural Integration (APU-X).

Problem: Constants like 77 MHz, 1400 W/m · K, and 40 ps gate delays lack citations.

Correction: Add a **Hardware Parameter Assumptions** table at the beginning of the hardware section, noting that values are typical for advanced 14nm processes and citing relevant literature benchmarks.

4.3 Issue 16: Definition of Quicksand Oja++

Location: Event-Driven Subspace Tracking.

Problem: The algorithm lacks a formal pseudocode definition.

Correction: Insert an **Algorithm Box** detailing the initialization, adaptive step-size computation (η_t^{eff}), noise estimation, and the orthogonalization step.

4.4 Issue 17: Reconcile “Shadow Worker” Terminology

Location: Internal Verification Completeness.

Problem: The term is used broadly without a strict definition.

Correction: Add a formal **Definition (Shadow-Worker Architecture)**. Define it as a dual-execution model with a primary execution path and a lightweight, parallel verification thread that reconstructs states via the inverse dictionary operator.

5 Implementation Checklist for Unified Thesis Compilation

Order	Issue	Severity	Status in Thesis	
1	Fix Sign Error in Euler-Lagrange Eq.	Critical	Pending	Compilation
2	Insert Full Truncated Margin Proof	Critical	Pending	Compilation
3	Correct Arithmetic Error (10^{-7} K)	Critical	Pending	Compilation
4	Standardize Asymptotic Bound to B^4/δ_k	High	Pending	Compilation
5	Unify Quantization Noise Tolerance	High	Pending	Compilation
6	Add Covering Number Lemma for $\mathcal{O}(\log t)$	High	Pending	Compilation
7	Rewrite GIM Necessity Proof	Medium	Pending	Compilation
8	Add Adaptive Step-Size Boundary Lemma	Medium	Pending	Compilation
9	Add Phonon-Precision Bridge Lemma	Medium	Pending	Compilation
10	Reconcile $L_{contract}$ Forms	Medium	Pending	Compilation
11	Clarify 95% Accuracy as Experimental Target	Medium	Pending	Compilation
12	Explicitly Transition CHS Ideal to Physical	Low	Pending	Compilation
13	Add MLRP Application Context	Low	Pending	Compilation
14	Change “Multiplicative” to “Additive”	Low	Pending	Compilation
15	Insert Hardware Parameter Assumptions Table	Low	Pending	Compilation
16	Define Quicksand Oja++ via Algorithm Box	Low	Pending	Compilation
17	Formally Define “Shadow Worker”	Low	Pending	Compilation